

May 15, 2026

**RE: Quantitative Research Analyst, Market Implementation Group**

Dear Hiring Committee,

I am writing to apply for the Quantitative Research Analyst role on your Quantitative Equity Market Implementation team in Vancouver. I am motivated by work that combines disciplined quantitative research, high-volume data analysis, and systematic improvement of production outcomes—and by the chance to grow into market implementation while contributing pipeline and research skills from day one.

At Royal Bank of Canada, I architected Logging as a Service (LAS), an institution-wide log repatriation platform built on Vector, Fluent Bit, Kafka, and Logstash that ingests 150 TB/day across 10,000+ systems. I defined OpenTelemetry log and metric schemas, enabled vendor-agnostic switching among Dynatrace, Datadog, and Splunk, and designed a 60 TB/day Microsoft 365 audit-log repatriation collector. I also drove OpenTelemetry standardisation—collection pipelines, schema contracts, and SDK guides adopted by 20+ engineering teams. This work is directly aligned with monitoring implementation outcomes, evaluating pipeline performance, and improving the research infrastructure behind large-scale systematic operations. Separately, I led 0-to-1 delivery of ASB Builder, a patent-pending, rules-based (not ML-based) enterprise platform mapping 1,000+ patterns to real-time decisions; digitalised RBC's technical reference model across 1,000+ technologies and 1,500+ capabilities; and led five engineers partnering with product, risk, and architecture stakeholders in Enterprise Architecture (Athena) certification workflows.

My background also includes rigorous research and applied ML. At Dalhousie I earned an MASc in Applied Computer Science (GPA 3.9/4.0), served as a research assistant on compiler/runtime call-stack algorithms, and taught graduate algorithms and cloud computing. At HealthCard I designed a Hyperledger-based insurance validation and trust-score system and deployed transformer-based document classification that cut manual review by 60%. At CryptoVantage I built real-time Ethereum streaming analytics with ML wallet clustering and risk scoring for 5AMLD compliance. Outside my day job, I developed a personal algorithmic trading system (live brokerage APIs, momentum/mean-reversion signals, automated risk limits) and a reinforcement-learning pricing agent (DQN/DDQN) that beat a simulation baseline by 18%; I also published peer-reviewed research on decentralised storage (Springer ICMR).

I am transparent that my professional experience is in enterprise data platforms and engineering rather than sell-side trading. I am applying because I want to apply my Python, statistics, pipeline, and research skills to systematic equity market implementation, and I learn quickly in collaborative environments. I communicate technical concepts clearly, welcome feedback, and would relocate to Vancouver for this opportunity.

Thank you for considering my application. I have enclosed my resume and can provide academic transcripts upon request. I would welcome a conversation about how I can support your Market Implementation team's research, monitoring, and cross-functional delivery goals.

Best regards,



Dhruv Doshi